

Vortex Compiler (VXC)

Language Specification and Phase 1 Lexer

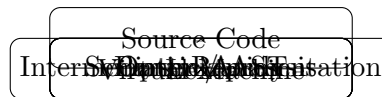
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1 Purpose

Vortex Compiler (VXC) is an educational compiler implemented in C. Its initial target is a custom stack-based virtual machine. Phase 1 defines the language and implements lexical analysis.

2 Compilation Pipeline



3 Lexical Specification

The lexer recognizes whitespace, line comments beginning with `//`, identifiers, decimal integer literals, keywords, and punctuation.

3.1 Keywords

`fn`, `let`, `int`, and `print` are reserved keywords.

3.2 Token Grammar

```
identifier  := letter (letter | digit | '_' ) *
integer     := digit +
letter      := 'A' .. 'Z' | 'a' .. 'z' | '_'
digit       := '0' .. '9'
```

4 Example Program

```
fn main() {
  let x: int = 10 + 20;
  print(x);
}
```

5 Phase 1 Implementation

The current C implementation reads a source file and prints each token with its type, lexeme, line, and column. This provides a testable foundation for the parser.

6 Build and Run

```
make  
./build/vxc examples/hello.vxc
```

7 Next Phase

Phase 2 will add a recursive-descent parser and an abstract syntax tree for declarations, expressions, and function bodies.